

Terms of Trade and Exchange Rate Regimes in Developing Countries

Christian Broda (2004)

Elyse Demkiw

Columbia University

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Theoretical Background: Demand Side

Chapters 10 & 13

Tradable consumption is a positive function of lifetime wealth:

$$C_t^T = C^T(r^*, TT_1Q_1, TT_2Q_2, (1+r_0)B_0)$$

Greater wealth raises demand for both tradables and non-tradables, pushing up their relative price:

$$p_t = \frac{1-\alpha}{\alpha} \frac{C_t^T}{F(h_1)} \quad (\text{demand schedule})$$

Using market clearing $C_t^N = Q_t^N = F(h_t)$:

$$p_t = p\left(C_t^T, h_t\right)_{+ -} \implies p_t = p\left(r^*, TT\right)_{- +}$$

Theoretical Background: Supply Side

Chapters 10 & 13

Firms choose h_t to maximise profits; imposing $Q_t^N = F(h_t)$:

$$\Pi_t = P_t^N F(h_t) - Wh_t$$

First-order condition: $P_t^N F'(h_t) = W$. Dividing by $P_t^T = \varepsilon_t P_t^{T*}$ (with $P_t^{T*} = 1$):

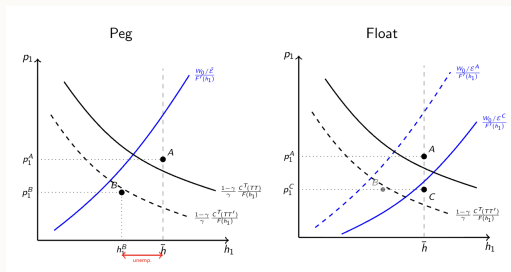
$$p_t = \frac{W/\varepsilon_t}{F'(h_t)} \quad (\text{supply schedule})$$

Peg vs. Float after a ToT Shock

$TT_t \downarrow \Rightarrow C_t^T \downarrow \Rightarrow p_t \downarrow$: demand schedule shifts left.

To maintain full employment at the new lower p_t , the real wage $W_t/\mathcal{E}_t$ must fall. With downward nominal wage rigidity, W_t cannot decrease

- ▷ **Float:** $\mathcal{E}_t \uparrow$ (depreciation) $\Rightarrow$ real wage falls $\Rightarrow$ supply shifts right $\Rightarrow$ full employment maintained
- ▷ **Peg:** $\mathcal{E}_t = \bar{\mathcal{E}} \Rightarrow$ supply frozen $\Rightarrow$ economy slides down supply curve $\Rightarrow h_1 < \bar{h}$, involuntary unemployment



This is the foundation of Friedman's hypothesis, Broda's contribution is quantifying this hypothesis by looking at ToT shocks and how they transmit to GDP, CPI and RER

$$GDP = Q_t^T + p_t F(h_t)$$

GDP moves through changes in h_t thus in pegs we expect a decline from a negative ToT shock while in floats we expect no change

$$CPI = (P^T)^\alpha (P^N)^{1-\alpha}$$

CPI moves through changes in price; in pegs P^T is fixed while non-tradable price would decline $\rightarrow$ fall in CPI, in floats P^N is stabilized via full employment while $P^T = \varepsilon P^{T*}$ increases $\rightarrow$ increase in CPI

Sample: 75 small open economies, non-oil-exporting, population $> 1\text{M}$, 1973–1996

Core identifying assumption: countries are too small to affect world prices

- ▷ **Market share test:** only 22 goods from 9 countries exceed 15% of world export share; these average just 6.2% of each country's total exports
- ▷ **Granger causality test:** exports, imports, and RER fail to Granger-cause ToT for developing countries

Exchange rate classification: Ghosh et al. (IMF) regime codes

Other specifications:

- ▷ Stability condition: regime must be identical and non-missing at $t - 2, \dots, t + 2$
- ▷ Country fixed effects via within-country demeaning

$$A_0 Y_{it} = A(L) Y_{it} + B(L) X_{it} + u_{it}$$

- ▷ Y_{it} : vector of ToT, real GDP, RER, price level for country i , year t
- ▷ A_0 : simultaneous interactions
- ▷ $A(L)$: autoregressive terms for endogenous vars; isolates the clean ToT shock
- ▷ $A(L)$: autoregressive terms for exogenous controls
- ▷ X_{it} : controls include financial development, financial openness, CA changes

Two main outputs:

1. **Impulse response functions:** trace GDP, RER, prices after a single ToT shock over time
2. **Variance decomposition:** share of GDP/RER/price fluctuations explained by ToT shocks

Broda's Findings

Responses to a 10% (PV) Permanent Fall in TT

Under Fixed Regimes

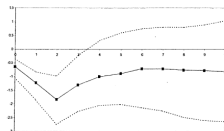


Figure 3: Real GDP Response

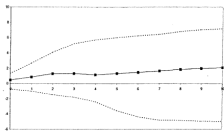


Figure 4: RER Response

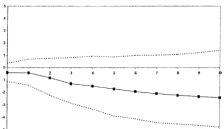


Figure 5: CPI Response

Under Flexible Regimes

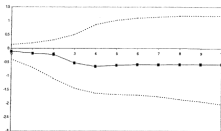


Figure 6: Real GDP Response

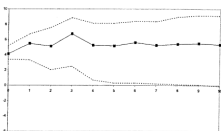


Figure 7: RER Response

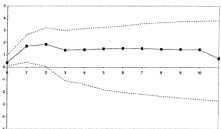


Figure 8: CPI Response

After 2 years of a negative ToT shock:

Fixed regimes

- ▷ GDP: -1.9%
- ▷ RER: 1.3%
- ▷ CPI -0.8

Flexible regimes

- ▷ GDP: -0.2%
- ▷ RER: 5.1% (immediate)
- ▷ CPI rises $\sim 2\%$

Differences vanish after ≈ 2 years. Wald tests reject equal ToT coefficients across regimes at 1% for both GDP and RER.

Key Variables

Variable	Source	Caveats	Fix
Tariff Data	WB Net Barter ToT; IMF CTOT (Gruss–Kebhaj)	CTOT ends 1995; poor fit for commodity-light countries	Granger test; robustness on 20 lowest- R^2 countries
Real GDP	PWT 10.01 rgdpna (2017 prices)	Annual; minor pre-1980 gaps	$\Delta \ln$; drop missing
CPI	WB WDI; IMF WEO Oct 2025	WDI gaps in 1970s; differing base years	Splice WEO onto WDI; $\Delta \ln$
RER	Darvas narrow REER; IMF REER	Broda's EREER_IX unavailable; bilateral low corr.	Compare $\Delta \ln$ across sources; Darvas for coverage
Regime	IRR coarse; LYS	Neither is Ghosh et al. more records, more floats	LYS robustness; IRR now standard
Trade Openness	WB WDI	Missing for select country-years	76% coverage

Robustness; Granger Causality on ToT

Test: do GDP, CPI, or RER Granger-cause ΔToT ? H_0 : no Granger causality

- ▷ **GDP** $\rightarrow$ **ToT**: $F = 0.44$, $p = 0.646$ fail to reject. *Broda*: $p = 0.98$ ✓
- ▷ **CPI** $\rightarrow$ **ToT**: $F = 0.62$, $p = 0.538$ fail to reject. *Broda's imports* $\rightarrow$ *ToT*: $p = 0.93$ ✓
- ▷ **RER** $\rightarrow$ **ToT**: $F = 2.80$, $p = 0.061$ marginal rejection at 10%. *Broda*: $p = 0.39$

If RER does affect ToT, the direction of bias depends on whether a country has monopoly power over its traded goods. Cross-checked with market share test

Difficulty with the RER

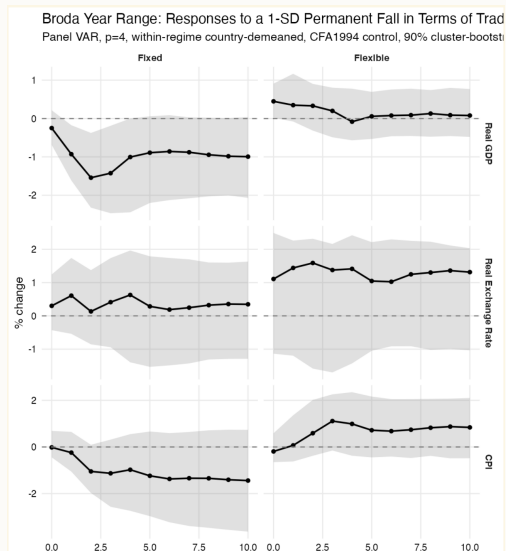
Broda's original series (EREER_IX, IMF) is unavailable. Alternatives compared:

Measure	Deflator	Closest to Broda?	Main Reproduction Risk
IMF REER (current)	CPI	Yes (same series)	Coverage gaps force bilateral fallback
Bilateral RER vs USD	CPI (dom. + US)	Broda's fallback	CPI revisions; misses non-USD shocks
Darvas REER	CPI	Methodologically yes	Different weights / vintages
Bruegel REER	CPI (some ULC)	Methodologically close	Mixed deflators; pre-1980 patchy

IMF REER and bilateral RER poorly correlated in $\Delta \ln$ for many sample countries, especially in the short run.

- ▷ Modern sources for RER, GDP, CPI are far different to what Broda used; Broda used data available in 2004, we have 20 years of hindsight, access to more accurate statistics & methods
- ▷ IMF REER and bilateral are not well correlated:
 - r difference > 0.75 : ZAF, CHL, GMB, COL, SLE, CMR, TTO
 - $0.5-0.75$: CAF, PRY, PNG, CRI, MWI, MEX, URY, GAB, CHN, NIC
 - < 0.5 : NZL, GHA, ISR, PHL, MYS, UGA, PAK, SGP, BDI, ZMB, DOM, BRA, BOL, MAR
 - log RER has better correlation than log differences, better long run correlation but they poorly track each other's short run changes

Replication Results



Fixed regimes

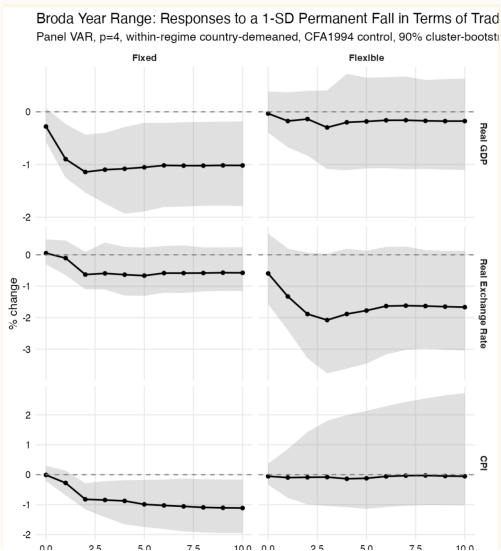
- ▷ GDP: -1.54% vs. -1.9%
- ▷ RER: 0.3% vs. 1.3%
- ▷ CPI: -1.05 more intense than Broda

Flexible regimes

- ▷ GDP: -0.3% vs. -0.2%
- ▷ RER: 1.59% vs. 5.1%
- ▷ CPI: $+0.95\%$ vs. $\sim 2\%$

GDP significant short-run; CPI significant short and long run; RER individually insignificant at 10% but peg-float gap is. Wald: better than Broda for CPI; $\approx 0.33\times$ his for RER

Extending the Sample to 2023



Anomalous results

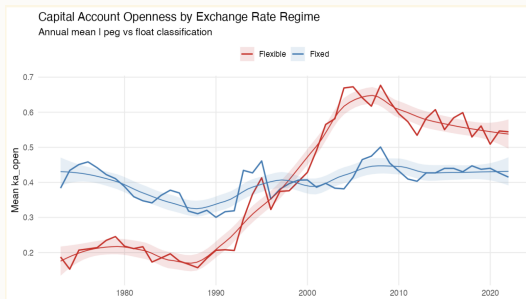
- ▷ GDP: consistent with Broda
- ▷ RER: massive *appreciation* for floats
- ▷ CPI: hovers around zero

Candidate explanations

- ▷ Transition to manufacturing: rerun market share test excluding CHN, POL, MEX, KOR, TUR
- ▷ Inflation targeting (17/73 countries) anchors CPI near zero
- ▷ Higher FX exposure and capital account openness

Capital Account Openness

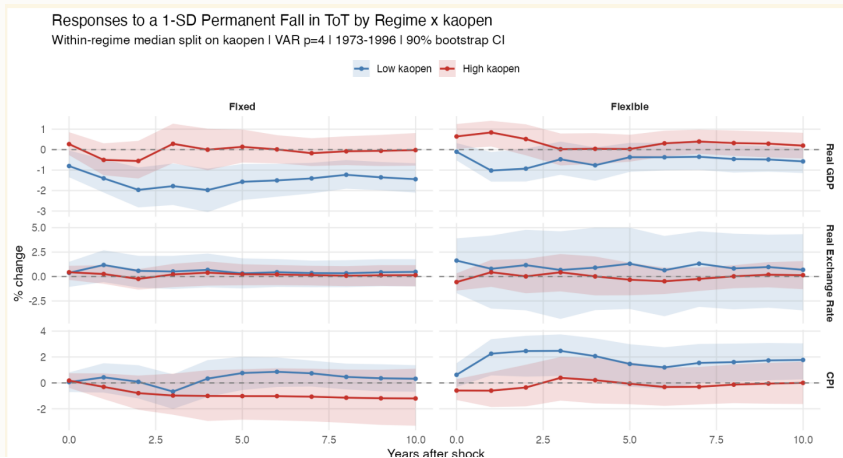
The baseline model assumes **free capital mobility**. Greater borrowing access lets countries smooth temporary ToT shocks, muting the short-run output response



Broda retroactively verifies unit root in 75% of panel ToT shocks, up to 25% could be temporary, and across all ToT shocks we don't know which were **perceived** as temporary; where capital account variation is relevant.

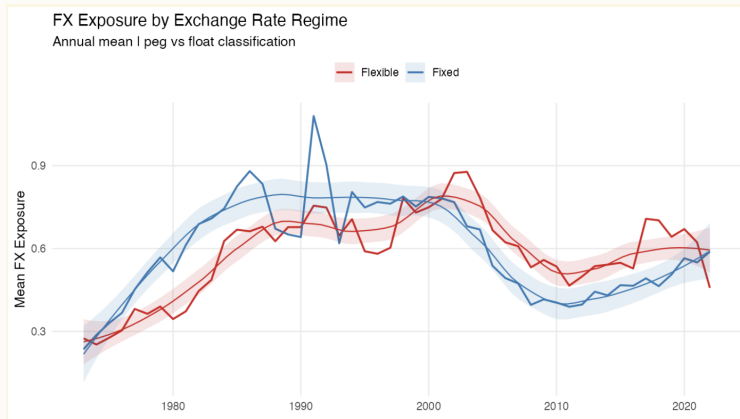
Capital Account Openness; Median Split

Sample split at the within-regime median of capital account openness; VAR + IRFs re-estimated separately



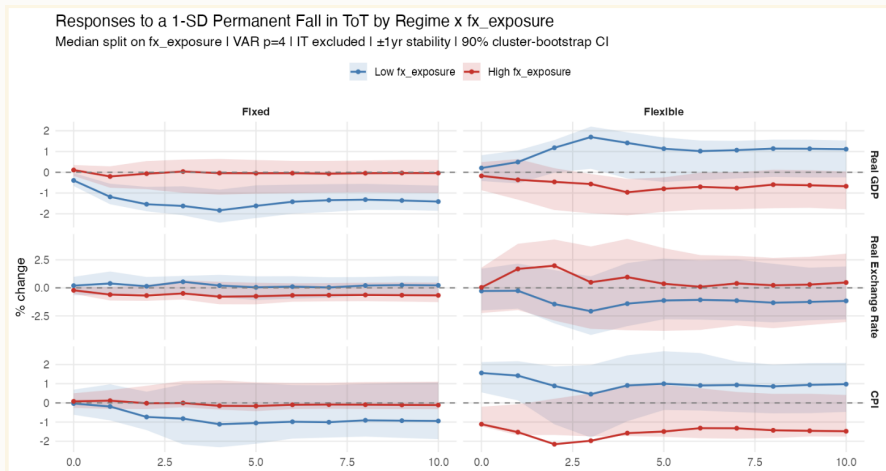
FX Exposure

Countries with high FX exposure $\left(\frac{\text{debt liabilities} - \text{FX reserves}}{\text{GDP}} \right)$ resist depreciation (it raises the domestic-currency cost of foreign debt)



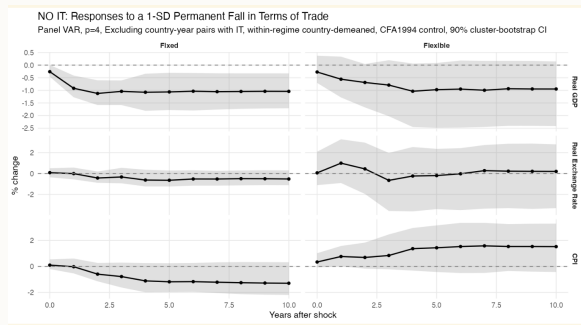
FX Exposure: IRF Split

Median split on FX exposure within each regime; inflation-targeting observations excluded



Inflation Targeting

Excluding the 17/73 inflation-targeting countries *reverses the sign* of the RER response in the extended sample.



Hypothesis: if $\alpha \uparrow$ (higher tradables share), depreciation disproportionately raises CPI rather than stabilising non-tradable prices

Extension: high-tradables IT central banks should undershoot their inflation target *less* after ToT shocks than low-tradables IT countries, since depreciation does more of the stabilisation work.